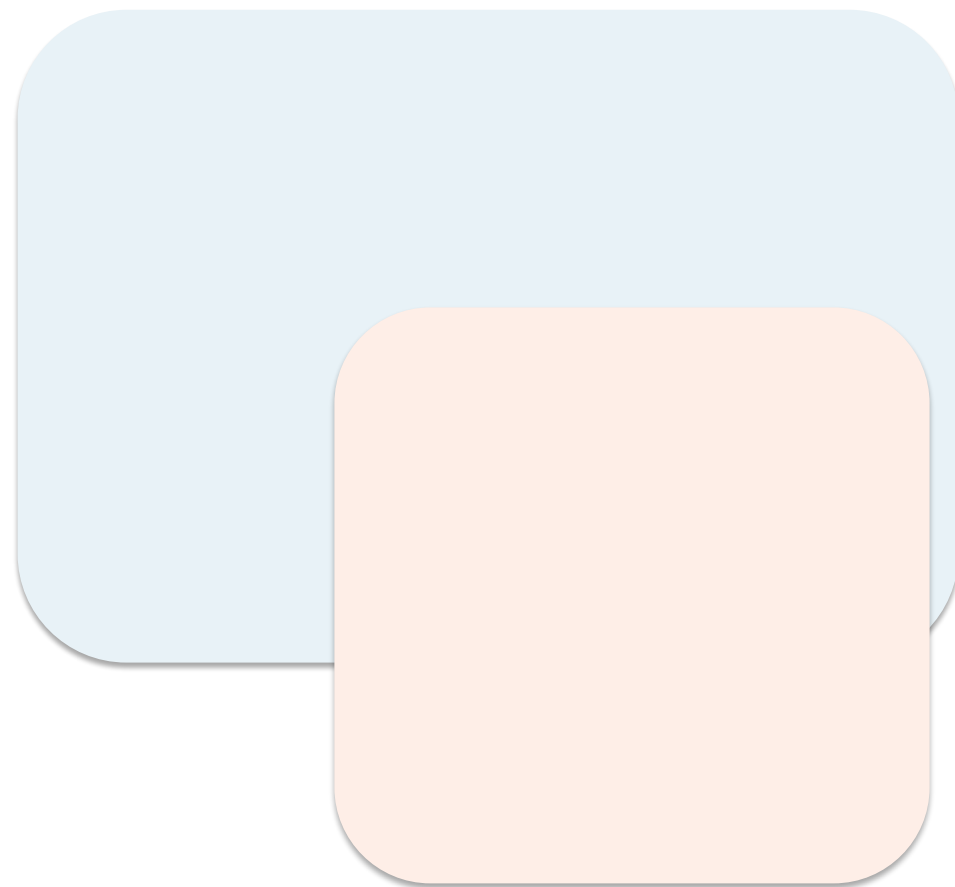




Quantum Computing and Genetic Algorithms for Software Testing

2026.07.17 Meeting

Participant: Justin, Anum



Paper 1 · Anomaly Detection in High-Energy Physics Using a Quantum Autoencoder

What does this paper do?

1 · Problem

Unknown new physics cannot be covered exhaustively by signal-specific supervised learning.

2 · Data

Simulated LHC collision events are reduced to limited physically meaningful features.

3 · Method

Angle encode the vector, train a variational QAE, and force trash qubits toward $|0\dots 0\rangle$.

4 · Result

QAE beats the paper's dense CAE baseline and is unusually data-efficient.

Core anomaly score

background-like event



trash fidelity $F \approx 1$
anomaly score $1-F \approx 0$



anomalous event
low F , high $1-F$

Searching for new physics without knowing its shape

Observed situation

No confirmed new interaction or particle has emerged from LHC data.

Limitation of supervised learning

A classifier must be trained on a specific simulated signal hypothesis.

Paper's response

Train only on Standard Model background and flag unfamiliar events.

Supervised

Make assumption of a signal model -> Labeled signal data->
Train background vs. signal

Model-agnostic (Unsupervised)

Train on background only ->
Learn its kinematic structure

High anomaly score means
“unlike background”

We just want to find anomalous part; anomaly doesn't stand for new physics.

How an autoencoder becomes an anomaly detector

II. CLASSICAL AUTOENCODERS

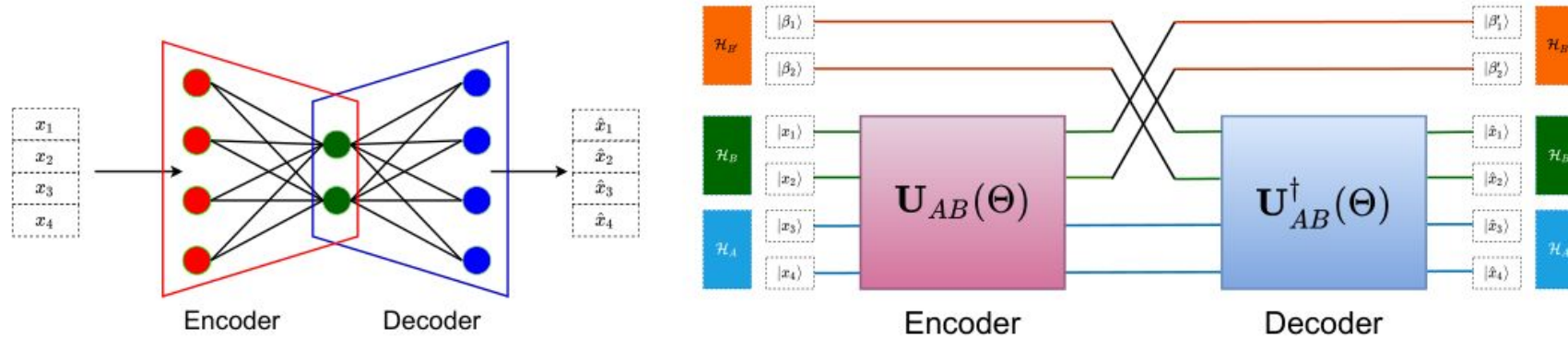


Figure 1 compares the CAE and QAE bottlenecks.

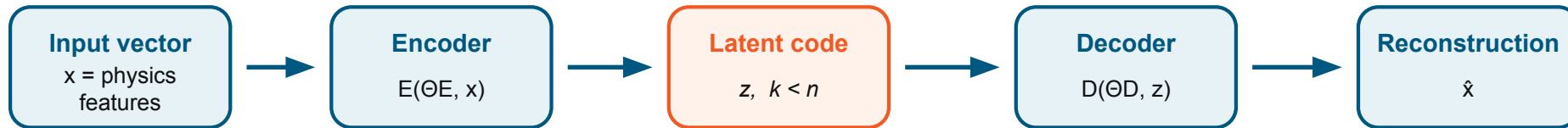
Classical autoencoder

$x \rightarrow \text{encoder} \rightarrow \text{low-dimensional latent } z \rightarrow \text{decoder} \rightarrow \text{reconstructed } \hat{x}$

Quantum complication

Unitary evolution preserves dimension, so a QAE must explicitly discard and replace trash qubits.

How the classical autoencoder actually works



$$\mathbf{z} = \mathbf{E}(\Theta_E, \mathbf{x}) \quad , \quad \hat{\mathbf{x}} = \mathbf{D}(\Theta_D, \mathbf{z})$$

Minimize reconstruction error RMSE

$$L(\mathbf{x}, \hat{\mathbf{x}}) = \sqrt{\frac{\sum_{i=1}^{i=n} (\hat{x}^i - x^i)^2}{n}}$$

1 Train on background only

The model learns the kinematic distribution of Standard Model events.

2 Background reconstructs well

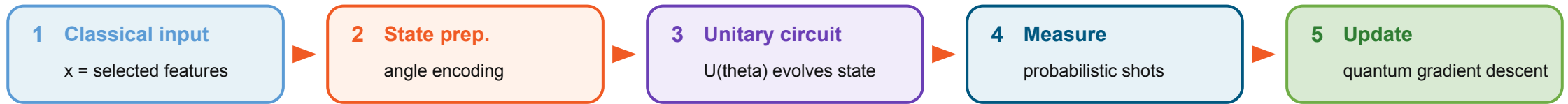
If $\mathbf{x}$ follows the learned distribution, $\hat{\mathbf{x}} \approx \mathbf{x}$ and the loss is small.

3 Signal-like events fail

A kinematically different event is reconstructed poorly, so loss becomes the anomaly score.

Many signals have a higher intrinsic dimension than background data due to their complexity

How the quantum autoencoder actually works



Angle encoding: one feature → one qubit

Real-valued observable ϕ_j becomes an R_x rotation angle on the Bloch sphere.

$$|\Phi\rangle = \bigotimes_{i=1}^n R_x(\phi_j) |0\rangle = \bigotimes_{j=1}^n \left(\cos \frac{\phi_j}{2} |0\rangle - i \sin \frac{\phi_j}{2} |1\rangle \right)$$

n-dimensional input needs n qubits.

Trainable evolution $U(\theta)$

After preparation, the parameterized circuit evolves the state:

$$|\Psi\rangle = U(\Theta) |\Phi\rangle$$

Input qubits are initially separable. CNOT gates create entanglement.

Measurement + training loop

Quantum measurement is probabilistic, so the same circuit is run many times.

shots → estimate observable → cost → update theta

We use **gradient descent** to update parameters theta, with **Fubini–Study metric** according to quantum-state geometry.

Step 1 — Prepare the quantum input state

Classical event vector

$$\mathbf{x} = \{p_T^{b1}, p_T^{l1}, p_T^{l2}, E_T\}$$

After rescaling: each component becomes an angle ϕ_j in $[0, \pi]$.

Angle encoding

$$|\Phi\rangle = \bigotimes_{i=1}^n R_x(\phi_j) |0\rangle = \bigotimes_{j=1}^n \left(\cos \frac{\phi_j}{2} |0\rangle - i \sin \frac{\phi_j}{2} |1\rangle \right)$$

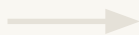
n input dimensions require n qubits. , 4 features -> 4 input qubits.

Physical meaning of the quantum state

q0

stores angle of
pT(b1)

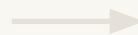
$R_x(\phi_1)|0\rangle$



q1

stores angle of
pT(l1)

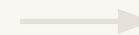
$R_x(\phi_2)|0\rangle$



q2

stores angle of
pT(l2)

$R_x(\phi_3)|0\rangle$



q3

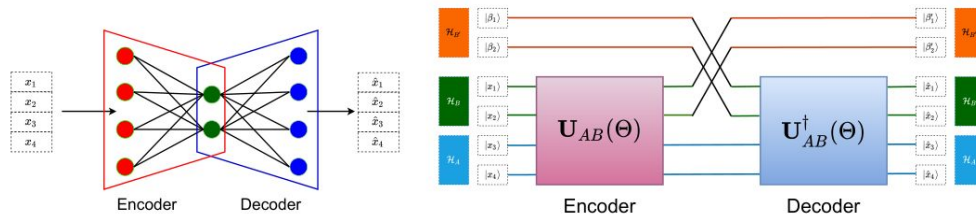
stores angle of
MET

$R_x(\phi_4)|0\rangle$

Immediately after angle encoding the state is separable: $|\Phi\rangle = |\phi_1\rangle \otimes |\phi_2\rangle \otimes |\phi_3\rangle \otimes |\phi_4\rangle$. CNOT gates are needed later to entangle features.

Step 2 — Build the bottleneck with latent, trash, and reference spaces

II. CLASSICAL AUTOENCODERS



Hilbert-space split

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_{B'}$$

A: latent qubits sent forward | B: trash qubits discarded | B': fixed reference space

Implementation steps in the paper

1

Prepare composite input

$$|\Phi\rangle_{AB} \otimes |\beta\rangle_{B'}$$

reference state beta is initialized as $|0\dots 0\rangle$

2

Encode only A+B

$$|\chi\rangle_{AB} \otimes |\beta\rangle_{B'} = (U_{AB}(\Theta) \otimes I_{B'}) |\Phi\rangle_{AB} \otimes |\beta\rangle_{B'}$$

$U_{AB}(\theta)$ is the trainable variational circuit

3

Swap trash with reference

$$|\chi'\rangle = I_A \otimes V_{BB'} |\chi\rangle_{AB} \otimes |\beta\rangle_{B'}$$

$V_{BB'}$ is the SWAP operation between B and B'

4

Optional decode

$$|\Psi\rangle = U_{AB}^\dagger(\Theta) \otimes I_{B'} |\chi'\rangle$$

decoder is the Hermitian conjugate of the encoder

Bottleneck: U is still dimension-preserving, so compression happens only because B is discarded and replaced by B'. If B already equals the reference, nothing important was thrown away.

Step 3 — Train the encoder by forcing trash qubits to become reference states

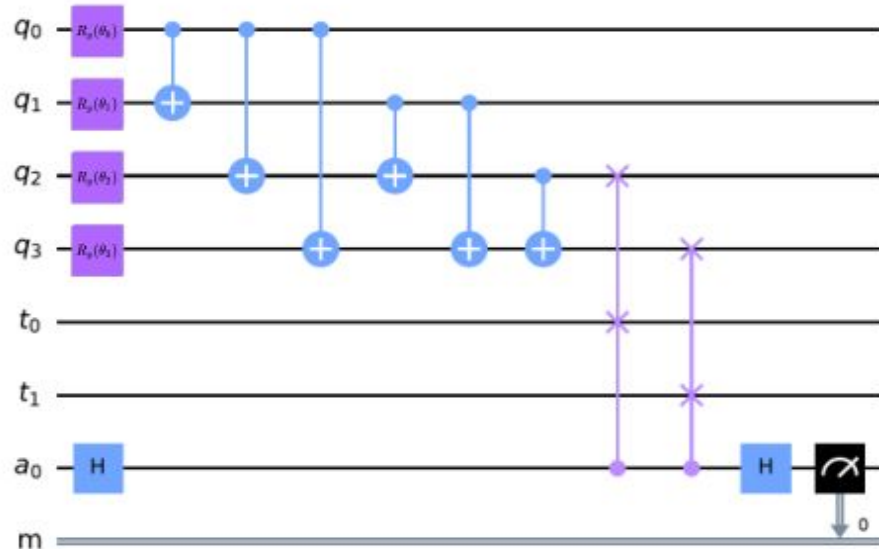


Figure 2 circuit: embedded input qubits $q_i \rightarrow$ trainable $R_y(\theta_i) \rightarrow$ CNOT entanglement $\rightarrow$ SWAP test against reference t_i and ancilla a_0 .

Perfect compression condition

$$U_{AB}|\Phi\rangle_{AB} = |\Phi^c\rangle_A \otimes |\beta\rangle_B$$

All useful information is in A; B contains no input information.

Fidelity target

$$F(|\beta\rangle_{B'}, \hat{\rho}'_B) = 1$$

$F = 1$ means the trash output is exactly the fixed reference.

Training cost function

$$C(\Theta) = - \sum_i p_i F(|\beta\rangle_{B'}, \hat{\rho}'_B)$$

Negative sign converts “maximize fidelity” into a minimization problem. The ensemble is needed to estimate the network output, not just as a classical mini-batch.

Anomaly score

$$s(x) = 1 - F$$

High fidelity $\rightarrow$ background-like. Low fidelity $\rightarrow$ hard to compress $\rightarrow$ anomalous.

Qubit resources

$$N_{\text{total}} = N_{\text{input}} + N_{\text{trash}} + 1$$

Example in Fig. 2: 4 input + 2 reference/trash + 1 ancilla = 7 total qubits.

End-to-end experimental workflow



Two benchmarks

Benchmark 1 · Main experiment

Background: $pp \rightarrow t\bar{t}$

Signal: $pp \rightarrow H \rightarrow t\bar{t}$

4 input variables $\rightarrow$ 4 qubits

Used for detailed training-size, ROC, significance, and hardware studies.

Benchmark 2 · Cross-check experiment

Background: $pp \rightarrow Z(\nu\bar{\nu})jj$

Signal: $pp \rightarrow hjj$, $h \rightarrow SS$ scalar dark-matter particle

6 input variables $\rightarrow$ 6 qubits

Used to test whether the QAE advantage carries to a different topology.

Representation changes



Experiment 1

Physics processes

Background: $pp \rightarrow t\bar{t}$
 Signal: $pp \rightarrow H \rightarrow t\bar{t}$
 $\sqrt{s} = 14 \text{ TeV}$
 $m_H = 1.0, 1.5, 2.0, 2.5 \text{ TeV}$

Event generation chain

MadGraph5_aMC@NLO $\rightarrow$ Pythia8 $\rightarrow$
 Delphes3 $\rightarrow$ FastJet
 Hard scattering $\rightarrow$ shower/hadronization $\rightarrow$
 detector $\rightarrow$ jets

Dataset size

Background $\approx 30\text{k}$
 10k train · 5k validation · 15k test
 Signal $\approx 15\text{k}$ per mass

Object cuts

anti- k_T jets, $R = 0.5$
 b-jets: $p_T^b > 30 \text{ GeV}$
 muons: $p_T^\mu > 30 \text{ GeV}$
 isolation $R = 0.5$

Decay and reconstruction logic

$$H \rightarrow t\bar{t}, \quad t \rightarrow bW, \quad W \rightarrow \mu\nu$$

The analysis focuses on the muon channel. The reconstructed b-jet, two muons, and missing transverse energy become the compact event description.

Final 4D vector passed to the scaler

$$x_{t\bar{t}} = \left[p_T^{b_1}, p_T^{\ell_1}, p_T^{\ell_2}, \cancel{E}_T \right]$$

$p_T^{b_1}$

leading b-jet transverse momentum

$p_T^{\ell_1}$

highest- p_T muon

$p_T^{\ell_2}$

second-highest- p_T muon

$\cancel{E}_T$

missing transverse energy from invisible recoil

Data simulation emphasis: raw detector information is not used directly. The paper first compresses each event into a small, physically interpretable vector before any quantum encoding.

Experiment 2

Physics processes

Background: $pp \rightarrow Z(\nu\bar{\nu})jj$

Signal: $pp \rightarrow hjj, \quad h \rightarrow SS$

The Higgs decays invisibly to scalar dark matter particles.

Generation and cuts

Same simulation chain as Experiment 1
 ≥ 2 jets with $p_T > 30$ GeV
 missing transverse momentum > 30 GeV

Dataset size

Background: 30k
 10k train · 5k validation · 15k test
 Signal: 15k test

Purpose

This benchmark checks that the QAE behavior is not specific to the $t\bar{t}$ final state.

Six-dimensional event vector

$$x_Z = [|\Delta\eta_{jj}|, m_{jj}, H_T^{1.0}, H_T^{1.5}, H_T^{2.0}, H_T^{2.5}]$$

$$H_T^{\eta_C} = \sum_{|\eta_i| < \eta_C} E_T^i \quad \eta_C \in \{1.0, 1.5, 2.0, 2.5\}$$

Mapping note: the paper again uses angular embedding. For this case the feature ranges are extended on the high side so that the non-periodic variables are not wrapped into equivalent angles.

Feature meanings

- $|\Delta\eta_{jj}|$: separation in pseudorapidity between the two leading jets
- m_{jj} : invariant mass of the dijet system
- $H_T^{\eta_C}$: transverse-energy sum inside a central pseudorapidity window
- The different η_C values probe how central or forward the transverse activity is

How the same QAE is trained and evaluated

Training target: make trash look like reference

$$|\beta\rangle_{B'} = |0\rangle^{\otimes N_{trash}}$$

$$C(\Theta) = - \sum_i p_i F(|\beta\rangle_{B'}, \hat{\rho}_{B'})$$

The negative sign converts “maximize trash fidelity” into a minimization problem.
A perfect reconstruction is possible when trash fidelity equals 1.

Testing target: anomaly score

$$s(x) = 1 - F(|\beta\rangle, \hat{\rho}_{trash})$$

Background-like

trash $\approx |0\dots 0\rangle$
 $F \approx 1$
 $s(x) \approx 0$

Anomalous

trash not close to reference
 F low
 $s(x)$ high

SWAP-test resources

$$N_{trash} \text{ is } N_{in} + N_{trash} + 1$$

Example: 4 input qubits + 2 reference/trash qubits + 1 ancilla = 7 total qubits.

Iteration loop



The paper trains and tests using only the encoder circuit, because trash fidelity acts as a proxy for reconstruction quality.

Optimization settings and classical baseline

QAE training schedule

Train events	Epochs	Updates
1	500	500
10	200	2,000
100	50	5,000
1,000	50	50,000
10,000	50	500,000

Quantum-gradient descent

$$\Theta_{i+1} = \Theta_i - \gamma \mathbf{G}^{-1}(\Theta_i) (\nabla_{\Theta} L(\Theta))_{\Theta=\Theta_i}$$

$\mathbf{G}$ is the Fubini–Study metric; the update respects quantum-state geometry.

$$G_{ij} = \text{Re} [\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle]$$

QAE fixed settings

5,000 shots per circuit evaluation
Weights updated for each data sample
Latent dimensions: 1, 2, 3

Classical autoencoder baseline

Dense CAE used for comparison

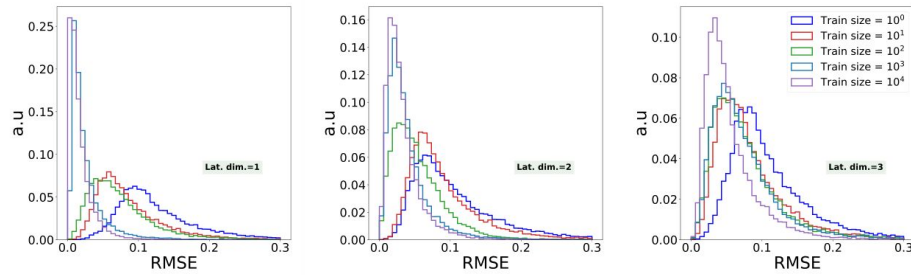
Encoder: input $\rightarrow 20 \rightarrow 15 \rightarrow 10 \rightarrow$ latent
Decoder: symmetric architecture
Hidden activation: ReLU; latent activation: Linear
Optimizer: Adam; learning rate: 10^{-3}
Loss: RMSE; batch size: 64
Parameters: $O(1000)$

Why this comparison matters

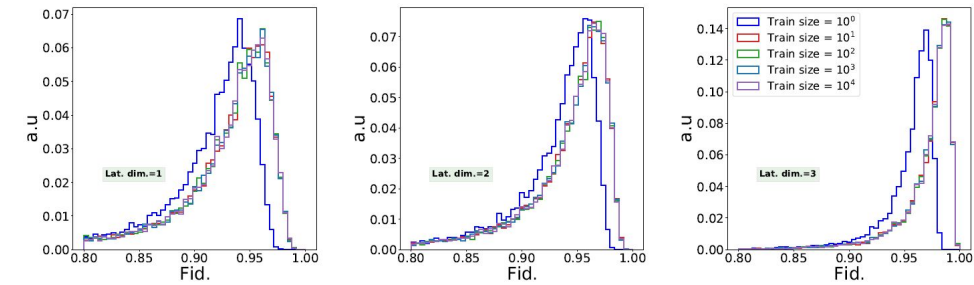
The QAE uses the same input variables as the CAE, but has only $O(10)$ trainable parameters. The paper is testing whether a shallow variational quantum circuit can learn a useful background compression with far fewer parameters.

Important caveat: fewer parameters and fewer epochs do not automatically mean lower runtime, because quantum evaluation requires repeated shots and circuit executions.

QAE learns from much smaller background samples



CAE: test RMSE distribution shifts strongly with more data



QAE: test fidelity stabilizes after about 10 events

Classical behavior

Efficient reconstruction needs roughly $\geq 10^3$ training samples; small samples show high uncertainty and poor generalization.

Quantum behavior

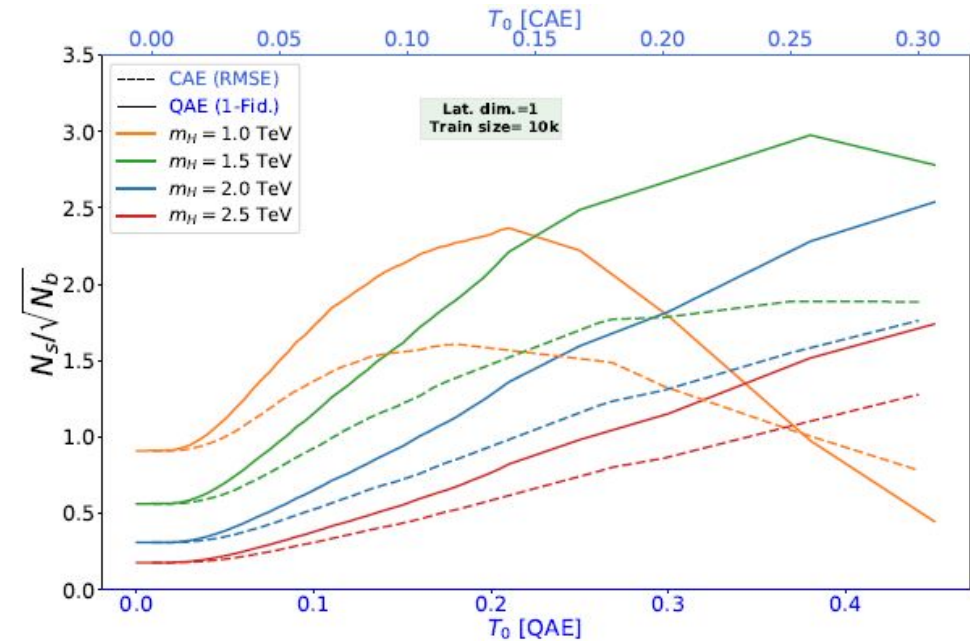
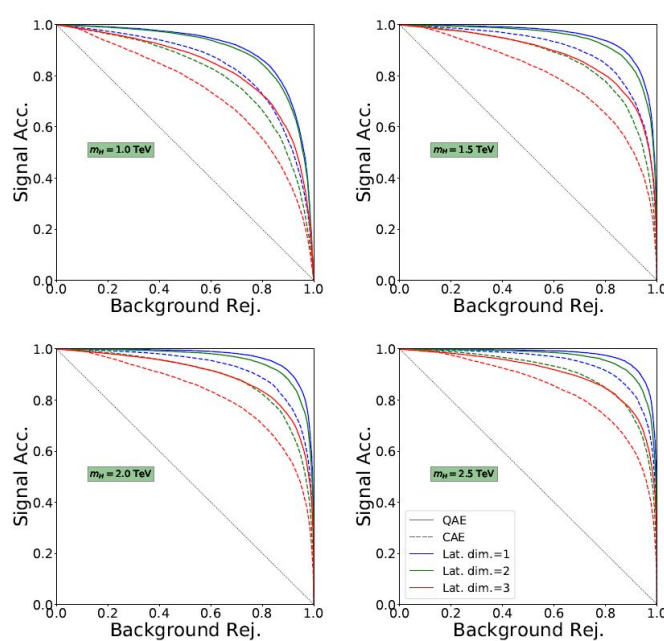
One training event does not fully converge, but from 10 events upward the fidelity distributions become nearly identical.

Interpretation

The authors suggest the simple circuit, measurement statistics, and quantum gradient descent may explain this small-sample efficiency.

This is the first major empirical claim: in the studied LHC benchmark, the QAE is much less dependent on training sample size than the dense CAE.

QAE beats the dense CAE on ROC and significance



ROC curves

For $m_H = 1.0, 1.5, 2.0$, and 2.5 TeV, QAE curves lie above CAE curves for all tested latent dimensions.

Bottleneck effect

The **smallest latent dimension** gives the strongest background rejection. Strong compression makes signal-like events harder to pass through.

Significance

Using 1-F for QAE and RMSE for CAE, the peak $N_S/\sqrt{N_B}$ is higher for QAE under $\sigma_S = 10$ fb and $L = 3000$ fb $^{-1}$.

The paper stresses that these metrics compare classifiers; the real LHC background rate remains much larger than the benchmark signal rate.

Hardware benchmark and second physical signature

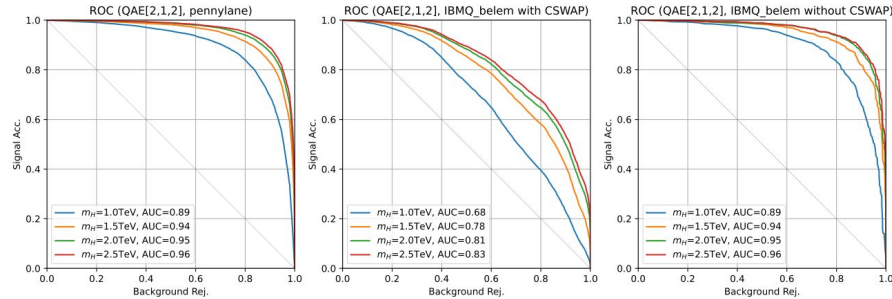
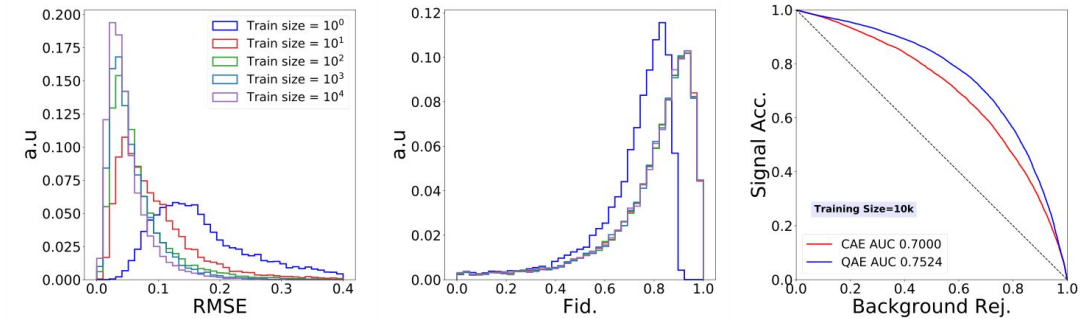


FIG. 9: ROC curves based on the fidelity distributions. Those evaluated by the **Pennylane** simulator (left panel), by the quantum device IBM-Q belem backend with the SWAP test (central panel), and with the second qubit measurements (right panel) are shown.



IBM-Q hardware lesson

The 2-input QAE is tested on IBMQ Belem. The SWAP-test version loses AUC because CSWAP is too deep and decoheres. Direct Pauli-Z trash measurement reproduces simulator AUC almost exactly.

Second benchmark lesson

For invisible Z background vs invisible Higgs signal, the QAE again converges with smaller data and outperforms the optimized dense CAE: QAE AUC = 0.7524 vs CAE AUC = 0.7000.

Together, these checks support the authors' claim that the behavior is not just a one-process artifact and that shallow measurement design is critical on NISQ devices.

Paper 2 · Applying Quantum Autoencoders for Time Series Anomaly Detection

What does this paper do?

1 · Research goal

Apply quantum autoencoders to time-series anomaly detection where training data contains only normal behavior.

2 · Data preparation

Use UCR univariate time series; cut each signal into 128-point sliding windows with stride 1.

3 · Encoding focus

Convert each 128-value window into a 7-qubit quantum state using amplitude encoding; use quasi-amplitude encoding on hardware.

4 · Experiments

Compare MSE reconstruction and Swap-Test latent analysis in simulation, then run a reduced setup on IBM Torino.

Core pipeline studied in the paper

raw time
series



128-point
window



7-qubit
amplitude
state



latent + trash
QAE



MSE / Swap
score

Research motivation and question

Problem setting

Anomalies are rare; many benchmarks provide normal training data and place the only anomaly in the test series.

Why time context matters

A value can look normal by itself but become anomalous within a local temporal pattern or over a longer segment.

Paper's goal

Evaluate whether QAE can detect anomalies in time windows and run on real NISQ hardware.

The paper studies two anomaly scores

Reconstruction error

Run encoder → reset trash → Encoder† and compute MSE between input and reconstructed window.

Latent representation analysis

Skip reconstruction and use the Swap-Test measurement: if trash differs from $|0\rangle$, the window compresses poorly.

Main research question: can a small QAE learn normal time-series structure from benign windows and separate unseen anomalous windows?

What data and tools did they use?

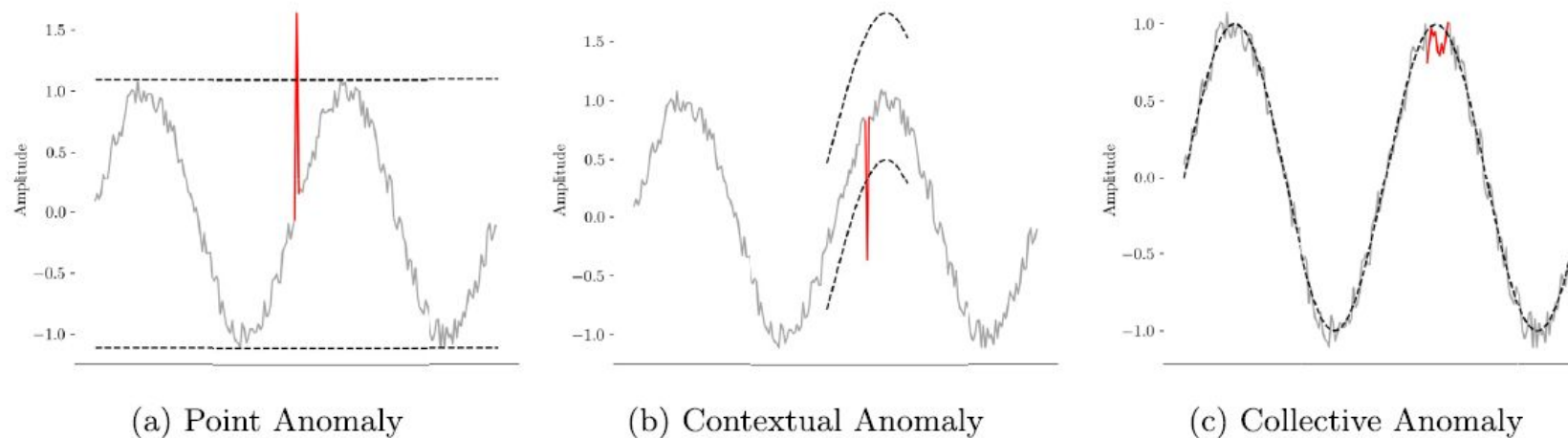


Figure 1: anomaly taxonomy; Fig. 4 in the paper shows the six selected UCR datasets and their red anomaly ranges.

Data source

UCR Time Series Anomaly Archive. Domains include medicine, sports, industry, robotics, space science, and more.

Datasets and tools

Simulator datasets: 28, 54, 99, 118, 138, 176.
Tools: PyTorch baseline; Qiskit 1.0.0 + Aer 0.13.1;
COBYLA; EfficientSU2 + IBM Torino hardware run.

Labels and split

Training series contains only benign values; each test series has a single anomaly range. Detection is valid if the selected window starts in the valid detection range.

Step 1 — Construct sliding windows

Raw univariate time series

$$X = \{x_0, x_1, \dots, x_T\}$$

Each x_i is one scalar value: a sensor reading, signal amplitude, physiological value, etc.

Window definition used by the paper

$$\mathbf{w}_i = [x_i, x_{i+1}, \dots, x_{i+127}]$$

window size

128 points
matches 2^7 amplitudes

stride

1
windows overlap heavily

input sample

one window
not one time point

count

$T-127$
windows from T points

Important detail: the paper does not normalize the raw series before windowing; normalization is performed during quantum state preparation.

Step 2 — Amplitude encoding into 7 qubits

For each 128-point window

$$\mathbf{w}_i = [w_{i0}, w_{i1}, \dots, w_{i,127}]$$

State preparation formula

$$|\phi_i\rangle = \frac{1}{\|\mathbf{w}_i\|} \sum_{j=0}^{127} w_{ij} |j\rangle$$

$$\alpha_j = \frac{w_{ij}}{\|\mathbf{w}_i\|}, \quad \sum_j |\alpha_j|^2 = 1$$

Why 7 qubits?

A K-qubit state has 2^K computational basis amplitudes. Since $2^7 = 128$, one full window fits into 7 qubits.

Basis-state mapping

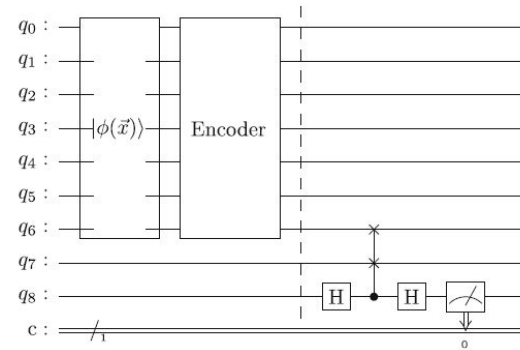
$w_{i0} \rightarrow$ amplitude of $|0000000\rangle$; $w_{i,127} \rightarrow$ amplitude of $|1111111\rangle$.

Important consequence

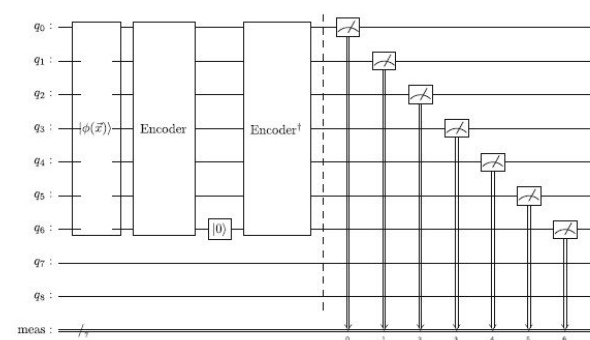
The vector norm is divided out. The state mainly preserves relative shape; absolute scale may be lost unless treated separately.

Amplitude encoding is qubit-efficient: N values require $\log_2(N)$ qubits. It is not necessarily hardware-efficient.

Step 3 — QAE encoder, latent qubits, and trash qubit



(a) Trainable quantum autoencoder architecture



(b) Trained quantum autoencoder architecture

7-qubit autoencoder

q_0 – q_5 are the 6-qubit latent space; q_6 is the trash qubit.

Training circuit

Training adds $q_7 = |0\rangle$ as reference and q_8 as Swap-Test ancilla, so the training circuit uses 9 qubits.

Compression target

After the encoder, q_6 should be close to $|0\rangle$, meaning all useful information has moved into q_0 – q_5 .

Decoder

For reconstruction, the decoder is $\text{Encoder}^\dagger$ after resetting q_6 . It is not trained separately.

Fig. 2: training circuit uses Swap-Test; final reconstruction circuit uses $\text{Encoder}^\dagger$ after resetting the trash qubit.

Data, ansätze, optimizer, and classical baseline

Data

Six UCR datasets: 28, 54, 99, 118, 138, 176. Window size = 128, stride = 1.

QAE size

7 data qubits; 9 training qubits; 6 latent qubits; 1 trash qubit.

Training

COBYLA optimizer, default settings, 45 iterations, 5,000 shots.

Execution

Qiskit 1.0.0 + Qiskit Aer 0.13.1; statevector simulator; 16 Intel Broadwell CPU cores; COBYLA optimizer.

Ansätze tested

- RealAmplitudes: alternating RY rotations and CX entanglement
- Entanglement patterns: circular, full, linear, SCA
- PauliTwoDesign: Pauli rotations + pairwise CZ gates
- Depth 2–10 → 21–77 trainable parameters

Classical baseline

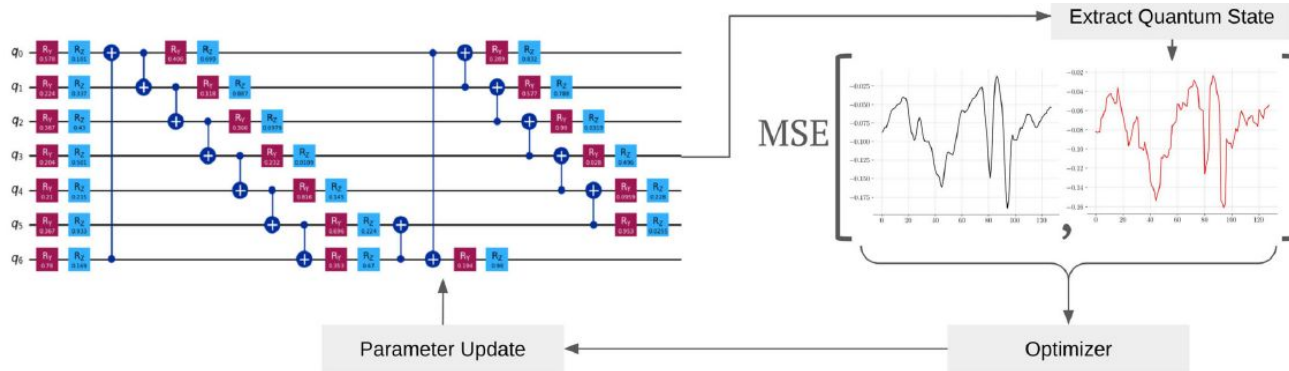
- LSTM autoencoder, input dimension = 128
- Latent dimension = 4
- Batch size = 150, Adam optimizer
- 250 epochs, 4,686 trainable parameters
- Anomaly score = reconstruction MSE

The paper averages each quantum ansatz configuration over 5 executions.

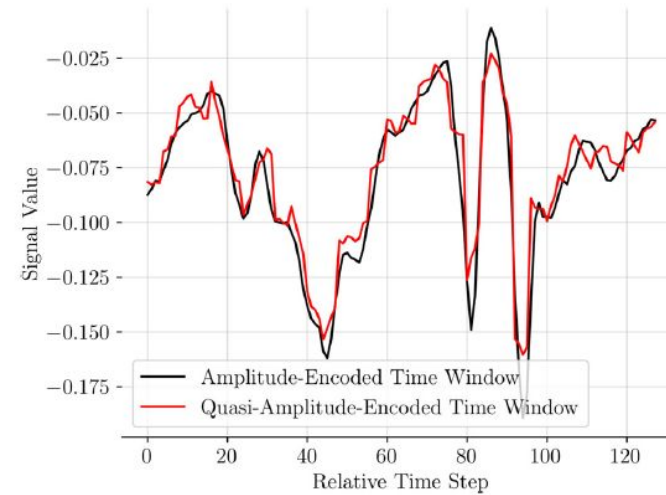
4 • QUASI-AMPLITUDE ENCODING

Exact amplitude encoding is replaced by a shallow variational state-prep circuit

How classical windows become hardware-ready quantum circuits



$$|\tilde{\phi}_i(\alpha_i)\rangle \approx |\phi_i\rangle, \quad \alpha_i^* = \arg \min_{\alpha} \text{MSE}(|\phi_i\rangle, |\tilde{\phi}_i(\alpha)\rangle)$$



Two parameter sets must not be confused

Input-specific α_i

EfficientSU2 parameters encode one particular window; fixed during QAE training.

Model parameters θ

PauliTwoDesign QAE encoder parameters are shared and optimized during training.

Quasi-encoding procedure

1. Target state: exact amplitude-encoded $|\phi_i\rangle$
2. Use EfficientSU2 with 7 qubits and 2 repetitions
3. Extract generated quantum state
4. Compute MSE to target state
5. SPSA updates state-prep parameters
6. Repeat 100 trials per window and keep the lowest-MSE circuit

QAE encoder training loop and optimizer

The model trains the encoder ansatz parameters θ

Optimizer

This paper uses COBYLA with default parameters, not gradient descent. COBYLA is derivative-free.

Training objective

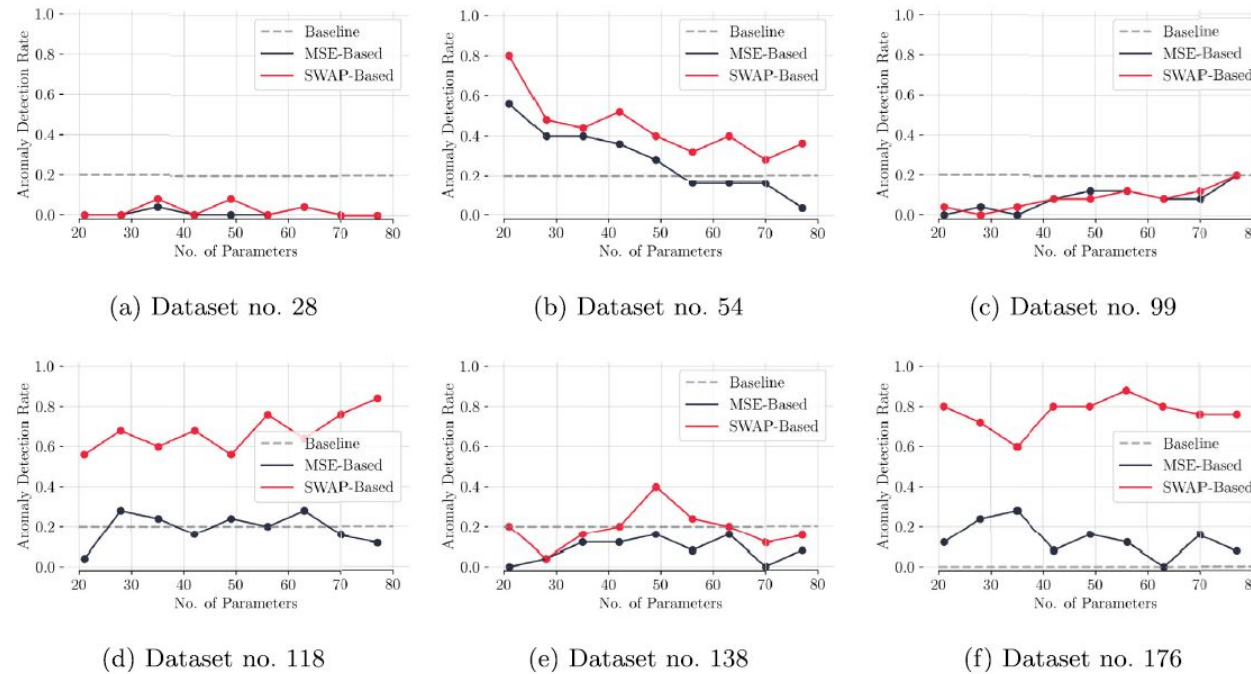
Minimize the probability that the Swap-Test ancilla is measured as 1.

$$L(\theta) = P_{\theta}(a_8 = 1) \approx \frac{\# \text{ measured 1s}}{N_{\text{shots}}}, \quad \theta^* = \arg \min_{\theta} L(\theta)$$

Hardware-specific settings

Dataset 54; 150 training windows; PauliTwoDesign with 2 repetitions; 21 trainable parameters; 45 epochs; 8,192 shots on IBM Torino.

Detection performance across ansätze and datasets



Key claim

For each dataset, at least one QAE configuration outperforms the LSTM baseline.

Important nuance

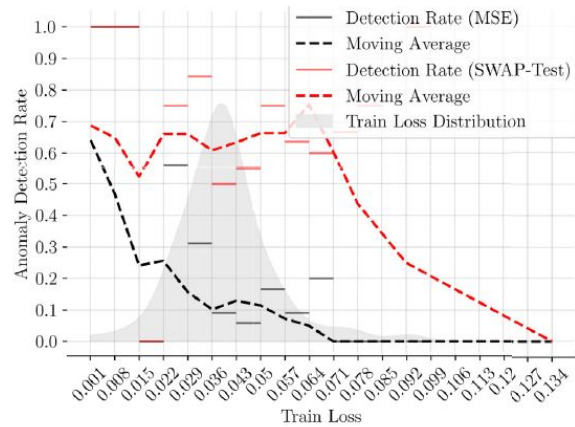
Not every ansatz works; many settings fail, especially on datasets 28 and 99.

Parameter efficiency

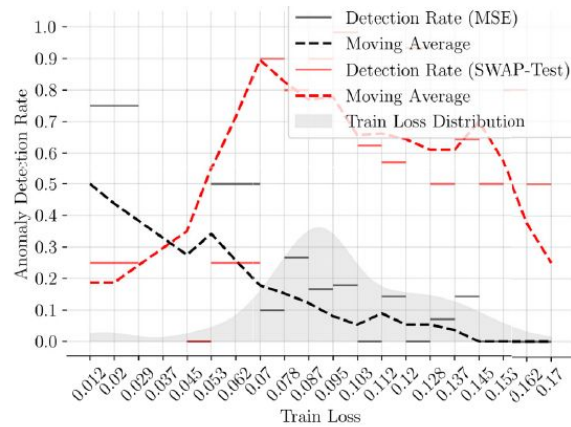
QAE: 21–77 parameters; 45 iterations, 250 epochs.
baseline: 4,686 parameters; QAE uses

Fig. 4 shows the six datasets. Fig. 11 compares average detection rates for MSE-based and Swap-based methods against the baseline.

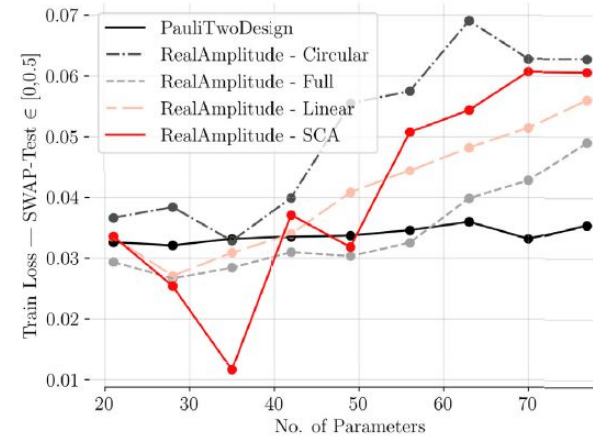
Swap-Test vs. MSE and training complexity



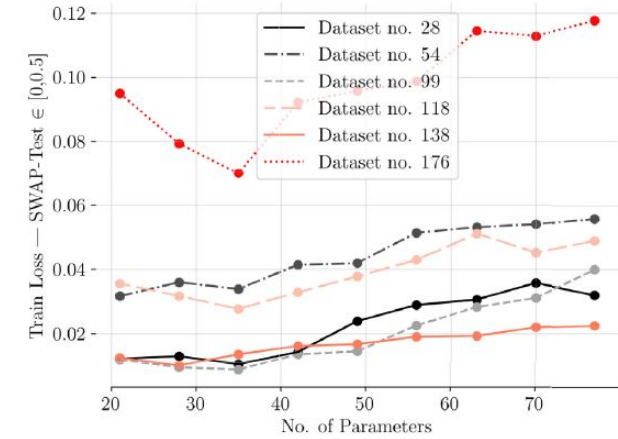
(a) Dataset no. 118



(b) Dataset no. 176



(a)



(b)

Swap-Test strength

More robust at higher training loss; especially helpful when anomalies are subtle and reconstruction noise is misleading.

MSE strength

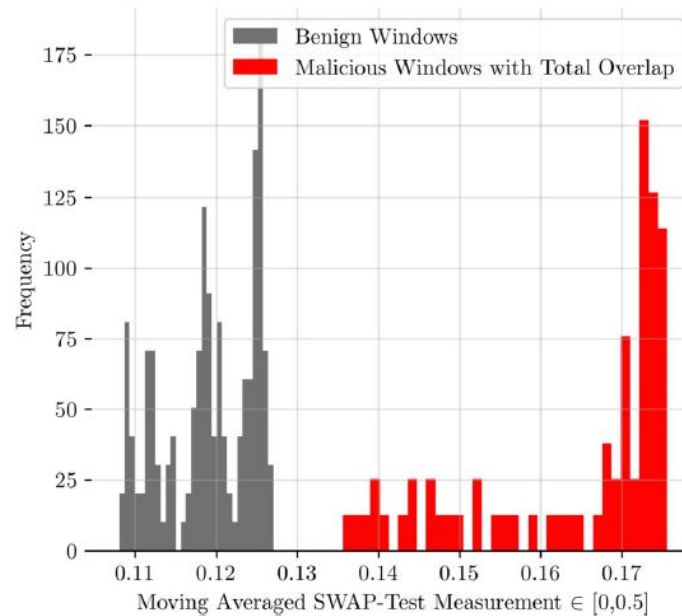
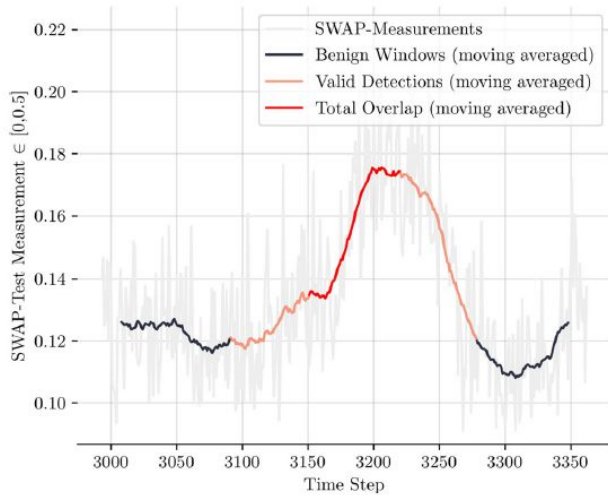
Can be better at very low loss, but requires extremely accurate reconstruction of all 128 values.

Ansatz complexity

More parameters often increase training loss with COBYLA; PauliTwoDesign remains more stable.

The paper's practical message: ansatz selection and scoring method are not secondary details—they determine whether the QAE detects anomalies.

Real quantum hardware results



Anomaly localization

Best IBM Torino QAE produces higher Swap-Test values where windows fully contain the anomaly.

Distribution separation

For fully anomalous windows: IBM Torino QAE overlap = 0; simulated QAE mean overlap ≈ 0.004 ; classical AE $\approx 70\%$.

Identification outcome

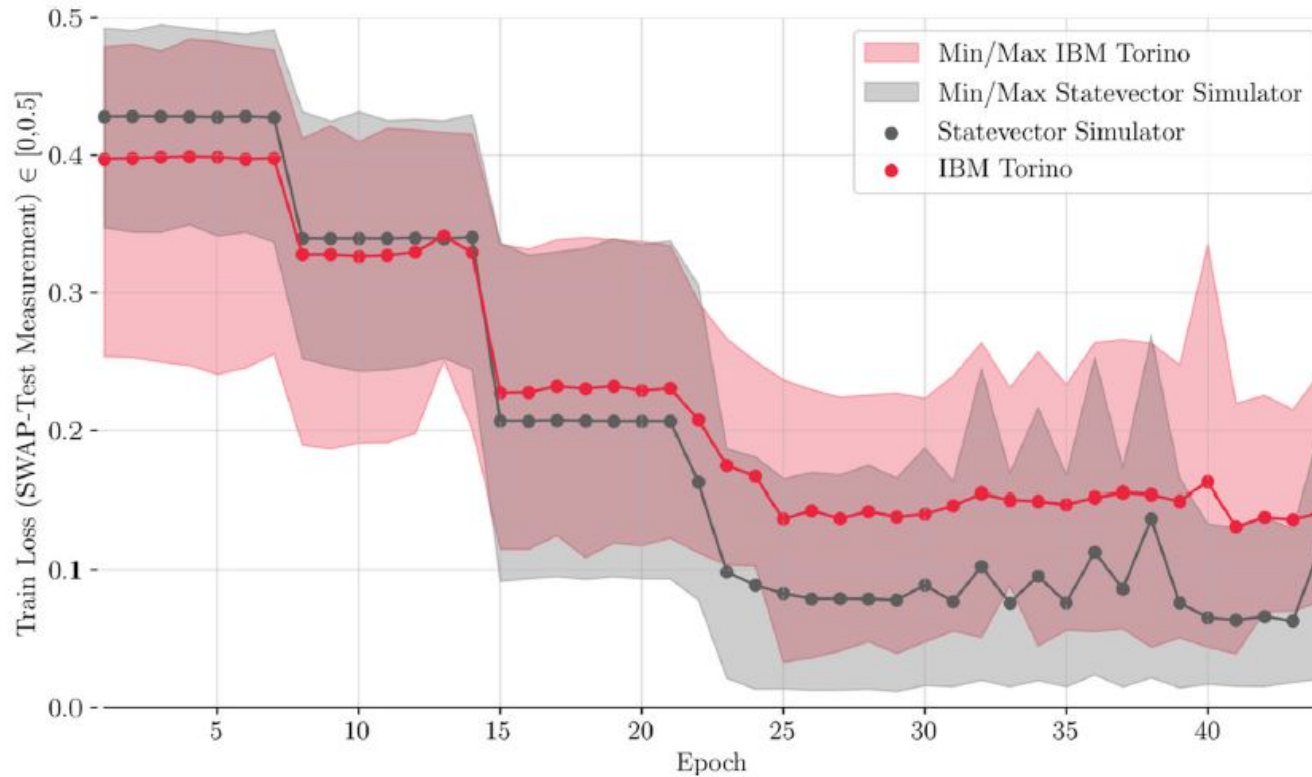
All tested QAE runs, simulator and hardware, correctly selected a window within the valid detection range; classical AE failed in Table 3.

Training metric

IBM hardware has larger loss than simulator, but separation of anomalous windows remains strong in this setup.

Fig. 14: moving-averaged Swap measurements rise inside the anomaly. Fig. 15: benign and fully anomalous distributions are separated.

Execution cost on real quantum hardware



Per-epoch time

Each epoch required ≈ 7 min of quantum computing time and ≈ 11 min elapsed time.

Per experiment

One real-hardware experiment consumed ≈ 5.25 h quantum time and took ≈ 8.25 h to complete.

Total study cost

The study used ≈ 21 h quantum computing time; pay-as-you-go estimate: \$120,960.

Interpretation

QAE uses fewer parameters, but current hardware execution is still expensive and not faster in practice.

Fig. 16 compares training loss on IBM Torino and the statevector simulator. Simulator loss falls faster; hardware loss is noisier but still converges.